

PAFA / PARR Platform

Implementation Guide

File Analysis · Entity Model · EF Core Configurations

Power BI Report Builder Export (Anonymised & Non-Anonymised)

Version 1.0 — June 2026

1. Source File Analysis & ETL Mapping

All 24 source Excel files have been inspected row-by-row. They fall into four structural patterns:

- Type A — Flat Long (grain: 1 row = 1 value per shipper × month): Read_Performance_by_Shipper_5, Shipper_Transfer_Read_Performance_5, Class_3_conversion*, Supply_Points_*
- Type B — Matrix Wide (months in columns, multi-row headers to skip): MOD520A all sheets, Report_1A/1B/2/3, 2B_21, Confirmed_Energy_Theft_*
- Type C — Multi-axis (EUC or Class in columns × Shipper in rows): Rpt_1364_PARR_AQ (8 sheets), AQ_at_Risk, EUC09_Reporting
- Type D — Transactional snapshot (GasDay × Shipper × EUC × LDZ — finest grain): 202604_Apr2026_SupplyPointCounts

1.1 File-by-File Parsing Instructions

Table below documents the exact skip-row count, column layout, unpivot strategy, and target MetricCode for every file.

Source File	Type	Sheet / Tab	Key Grain	ETL Action
MOD520A Non-Anon (2B.x)	Matrix Wide	Estimated & Check Reads, Read Performance, etc.	Shipper × Month	Skip rows 1–2; row 3 = months header; unpivot cols → long; split PC sub-tables
MOD520A Anonymised (2A.x)	Matrix Wide	Same sheets, anonymised labels	Shipper × Month	Identical transform; map AnonymizedLabel via Dim_Shipper
Read_Performance_by_Shipper_5	Flat Long	Data	Shipper × Topic	Direct load; map Topic → MetricCode (AQ_READ_PERF_PCT); row 1 = header
Shipper_Transfer_Read_Performance_5	Flat Long	Data	Shipper	Direct load; 3 columns (Count, Pct, ShortCode) → MetricCode TRANSFER_COUNT / TRANSFER_READ_PCT
AQ_at_Risk_Mar_2026	Wide (classes in cols)	AQ At Risk GWh PAFA, Percentage Overdue PAFA	Shipper × Class	Row 1 = class headers; unpivot → AQ_AT_RISK_GWH + AQ_AT_RISK_PCT per class
Class_3_conversion AQ Count/Pct	Flat Long	Data	YearMonth × ShortCode	Row 1 = header; direct load → CLASS3_CONV_COUNT + CLASS3_CONV_AQ + CLASS3_CONV_PCT
Supply_Points_with_Min_Pct PC2/PC3	Flat Long	Data	YearMonth × ShortCode	Direct load → MIN_PCT_REQ_PCT / MIN_PCT_NOT_MET_COUNT / MIN_PCT_NOT_MET_AQ

Source File	Type	Sheet / Tab	Key Grain	ETL Action
Report_1A Vacant within month	Matrix Wide	Results	Shipper × YearMonth	Row 1 blank, row 2 = periods, row 3 = label; skip rows 1–4, row 8 starts shipper data; unpivot → VACANT_IN_MONTH_COUNT
Report_1B Vacant end of month	Matrix Wide	Results	Shipper × YearMonth	Same as 1A → VACANT_EOD_COUNT
Report_2 Pct MPRN Must Read age	Multi-header Wide	Results	Shipper × Month × AgeBucket	Rows 1–3 multi-header; col pairs per month (a. 0–6 / b. 6–12 / c. 12+); unpivot → MUST_READ_REMOVED_AGE_PCT
Report_3 Count MPRN Must Read	Matrix Wide	Results	Shipper × Month	Row 1–2 header; unpivot → MPRN_ENTERING_MUST_READ_COUNT
2B_21 Corrective Opening Rejections	Matrix Wide (3 metrics/month)	Sheet1	Shipper × Month	Row 1 = months; row 2 = 3 metrics per month (Count, %Recv, %Raised); unpivot triplets → COMR_COUNT / COMR_REJECT_RECV_PCT / COMR_REJECT_RAISED_PCT
Confirmed_Energy_Theft Claim/WD P41+P106	Multi-header Wide	Sheet1	Shipper × Month × Metric	Rows 1–4 multi-header; pair of cols per month (%Claims, %Energy); 4 files → 4 MetricCodes
Rpt_1364_PARR_AQ (8 sheets)	Multi-axes EUC × Shipper	Report 1–8	Shipper × EUC	Each sheet = 1 metric; row 1 = title; row 2 = EUC cols; unpivot EUC → AQ_CALC_PCT etc.
EUC09_Reporting_PAC	Multi-header Wide	MET and NOT MET, Reclassified By	Class × Shipper × Month	Rows 1–4 header; row 5 = months; unpivot → CLASS1_ABOVE_COUNT + RECLASSIFIED_COUNT
202604_Apr2026_SupplyPointCounts	Transactional Long	Report 2	GasDay × Shipper × Class × EUC × LDZ	Row 3 = real header; rows 4+ = data; direct load → Fact_SupplyPointSnapshot

1.2 Complete MetricCode Catalogue

Every MetricCode that the ETL will write to the metric_values table (or specialised fact tables). The Unit column drives axis formatting in Power BI Report Builder RDL files.

MetricCode	Report Code	Source File	Unit	ETL Note
EST_PCT	2A.1	MOD520A Est & Check Reads	Pct	Row 3 months; col "Est" sub-header; unpivot; one row per shipper/month/class
CHECK_COUNT	2A.1	MOD520A Est & Check Reads	Count	Col "Check" sub-header in same sheet; same unpivot
NO_METER_PCT	2A.2	MOD520A No Meter Recorded in SP	Pct	Unpivot 12 months; ClassId from sub-table row offset
NO_METER_FLOW_PCT	2A.3	MOD520A No Meter + Data Flows	Pct	Same as 2A.2 but different sheet

MetricCode	Report Code	Source File	Unit	ETL Note
TRANSFER_READ_PCT	2A.4	Shipper_Transfer_Read_Perf_5	Pct	Column "Transfer Read Performance" → direct load
TRANSFER_COUNT	2A.4	Shipper_Transfer_Read_Perf_5	Count	Column "# Transfers" → direct load
READ_PERF_PCT	2A.5	MOD520A Read Performance	Pct	Row 3 = PC1/PC2/PC3/PC4-Monthly/PC4-Annual; map each col to ClassId
MRE01026..MRE01030	2A.6	MOD520A Meter Read Validity	Pct	5 MRE sub-tables; MetricCode = MRECode from Dim_LookupValue
NO_READ_1YR..4YR	2A.7	MOD520A No Read 1,2,3,4 yrs	Pct	4 sheets (Class 1–4); unpivot EUC cols × YearBand rows
AQ_CORR_REASON_01..09	2A.8	MOD520A AQ Correction by Reason	Count	→ Fact_AqCorrectionByReason; ReasonCodeLookupId FK
STD_CF_COUNT	2A.9	MOD520A Standard Correction Factors	Count	Unpivot EUC cols; store EucCode on MetricValue
REPLACED_READ_COUNT	2A.10	MOD520A Replaced Meter Reads	Count	Same as 2A.9; different sheet
CLASS1_ABOVE_COUNT	2A.11a	EUC09_Reporting MET and NOT MET	Count	Rows 5+; months in cols from col 3; forward-fill CurrentClass col
CLASS1_ABOVE_AQ	2A.11a	EUC09_Reporting MET and NOT MET	GWh	Second value column per month in same sheet
RECLASSIFIED_COUNT	2A.11b	EUC09_Reporting Reclassified By	Count	Second sheet; same unpivot logic
AQ_READ_PERF_PCT	2A.12a/b/c	Read_Performance_by_Shipper_5	Pct	Column 1 = value; col 2 = Topic (→ ObligationType Lookup); col 3 = ShortCode
AQ_AT_RISK_GWH	2A.13	AQ_at_Risk GWh sheet	GWh	Row 1 = class headers (Class1..Class4 variants); unpivot → ClassId FK
AQ_AT_RISK_PCT	2A.13	AQ_at_Risk Pct sheet	Pct	Second sheet; same logic
THEFT_CLAIM_OBJ_PCT	2A.14	Confirmed_Energy_Theft_Claim P41	Pct	Rows 3–4 = paired header (month + metric); unpivot per month pair
THEFT_CLAIM_ENERGY_PCT	2A.14	Confirmed_Energy_Theft_Claim P41	Pct	Second column of each month pair
THEFT_WD_OBJ_PCT	2A.14	Confirmed_Energy_Theft_WD P41	Pct	Withdrawal file; same structure
THEFT_P106_*	2A.14	Confirmed_Energy_Theft_* P106	Pct	Same 4 metrics as P41 files; store Period flag in LookupValue
CLASS3_CONV_COUNT	2A.15	Class_3_conversion AQ Count	Count	Direct long load; YearMonth col → PeriodId; SRVC_PRVDR_CD → ShipperShortCode
CLASS3_CONV_AQ	2A.15	Class_3_conversion AQ Count	GWh	"Reclassified Rolling AQ" col
CLASS3_CONV_PCT	2A.15	Class_3_conversion Pct file	Pct	Second file; same structure → PCT column

MetricCode	Report Code	Source File	Unit	ETL Note
MIN_PCT_REQ_PCT	2A.16	Supply_Points_Min_Pct PC2/PC3	Pct	Direct long load; two files (PC2, PC3) → ClassId = 2 or 3
MIN_PCT_NOT_MET_COUNT	2A.16	Supply_Points_Min_Pct Not_Met	Count	Companion "not met" files → same ClassId split
MPRN_REMOVED_PCT	2A.17	Report_1 Pct MPRN Must Read	Pct	Matrix wide; periods in row 2; skip rows 1–4; unpivot
MUST_READ_AGE_PCT	2A.17	Report_2 Must Read age bucket	Pct	Triple header rows; AgeBucket from LookupValue (a/b/c); col pairs per month
MPRN_ENTERING_COUNT	2A.17	Report_3 Count MPRN Must Read	Count	Matrix; row 1–2 header; unpivot
COMR_COUNT	2A.18	2B_21 Corrective Opening Rejections	Count	Row 1 = months; row 2 = 3-col groups; column 0 of each group
COMR_REJECT_RECV_PCT	2A.18	2B_21 Corrective Opening Rejections	Pct	Column 1 of each 3-col group
COMR_REJECT_RAISED_PCT	2A.18	2B_21 Corrective Opening Rejections	Pct	Column 2 of each 3-col group
VACANT_IN_MONTH_COUNT	2A.19	Report_1A Vacant within month	Count	Skip rows 1–7 (totals at row 5); data from row 8; unpivot periods
VACANT_EOD_COUNT	2A.19	Report_1B Vacant end of month	Count	Same structure; → VACANT_EOD_COUNT
AQ_CALC_PCT	2B.11	Rpt_1364 Report 1	Pct	EUC in cols; first metric sheet
AQ_INC_PCT	2B.11	Rpt_1364 Report 2	Pct	AQ increases; unpivot EUC
AQ_DEC_PCT	2B.11	Rpt_1364 Report 3	Pct	AQ decreases
AQ_CALC_FREQ	2B.11	Rpt_1364 Reports 4–7	Count	4 frequency distribution sheets
AQ_FAIL_COUNT	2B.11	Rpt_1364 Report 8	Count	Failure counts per shipper × EUC

1.3 Anonymous ⇄ Non-Anonymous Mapping

The **Dim_Shipper.AnonymisedLabel** column is the sole mechanism for switching between Schedule 2A (industry peer, city-name labels) and Schedule 2B (PAC, real company names). All stored procedures receive `@AnonMode bit` and apply a `CASE WHEN @AnonMode=1 THEN s.AnonymisedLabel ELSE s.FullName END AS ShipperDisplay` expression. No separate tables, RLS rules, or duplicate datasets are needed.

ShortCode	FullName (2B / Non-Anon)	AnonymisedLabel (2A / Anon)	Validation Status
AGA	Total Energies Gas & Power Limited	Rome	Confirmed
BRK	Brook Green Trading Ltd	Manama	Confirmed
BUS	British Gas Trading Limited	Brazzaville	Confirmed
NGS	SEFE Energy Limited	Gitega	Confirmed
GLC	Corona Energy Retail 4 Limited	Papeete	Confirmed

ShortCode	FullName (2B / Non-Anon)	AnonymisedLabel (2A / Anon)	Validation Status
SHE	SSE Energy Supply Ltd	Washington	Confirmed
PSL	P3P Energy Supply Limited	Bangui	Confirmed
EUK	Axpo UK Limited	Valletta	Confirmed
NGD	NPower Commercial Gas Ltd	Thimphu	Confirmed
SBP	Sembcorp Utilities (UK) Limited	Canberra	Confirmed
VOL	ENGIE Gas Shipper Limited	Philipsburg	Confirmed
HUD	Shell Energy UK Limited	Lisbon	Confirmed
SOG	EDF Energy Customers PLC	Taipei	Confirmed
BSH	Barrow Shipping Limited	Marigot	Confirmed
MRB	Marble Power Ltd	Warsaw	Confirmed
RWE	RWE Supply & Trading GMBH	Tehran	Confirmed
TET	Ceres Energy Limited	Sarajevo	Confirmed
VEC	Corona Energy Retail 2 Limited	Ankara	Needs CDSP validation
DRA	Drax Power Limited	Khartoum	Needs CDSP validation
FLX	Flexitricity Limited	Monaco	Needs CDSP validation
HEP	Hartree Partners Power & Gas Co Ltd	Luanda	Needs CDSP validation

2. Entity Model & EF Core Configurations

The data model uses a **star-schema generic pattern** centred on the `metric_values` table (equivalent to `Fact_KpiValue` in the technical plan). Two specialised fact tables handle the outlier formats (AQ corrections by reason code and the fine-grain supply point snapshot). All dimension data is seeded at startup via the DataSeeder.

2.1 Entity & Table Reference

Legend: Rows with orange background = MISSING in current project and must be created. Rows with green background = already coded but require additions.

Entity / Table	Key Columns	Maps To	Why It Exists
Shipper	Id, ShortCode, Name, AnonymisedLabel, IsActive	Dim_Shipper	Master shipper reference. AnonymisedLabel drives the toggle between 2A (city names) and 2B (real names) in every stored procedure.
ShipperAlias	Id, ShipperId, AliasCode, ValidFrom, ValidTo	shipperAlias	Historical short-code changes (e.g. NGS → SEFE). ETL resolves past codes via ValidFrom/ValidTo before writing MetricValues.
ReportingPeriod	PeriodId (YYYYMM), Year, Month, MonthLabel	Dim_Period	Natural-key integer avoids date arithmetic; enables BETWEEN 202403 AND 202604 range queries. Seeded 2023–2028.
ProductClass	Id (1–4), Code (PC1–PC4), AQThresholdLow, MinReadPercentage	product_classes	Static lookup: PC1 ≥ 732 MWh, PC2 medium NDM, PC3 small NDM WAR, PC4 IGT Small. Used as ClassId FK in MetricValue.
EucBand	EucCode (EUC01–09), Description, AqThresholdMinKWh, AqThresholdMaxKWh	euc_bands	EUC (End-User Category) lookup. Required dimension for reports 2A.7, 2A.9, 2A.10, Rpt_1364.
ReportDefinition	ReportCode (2A.1–2A.19), Topic, SplitBy, ReportFormat, RollingMonths, DisplayOrder	report_definitions	Source of truth for every stored procedure parameter. Power BI Report Builder reads this table to label axes and apply rolling window.
MetricDefinition	MetricCode, ReportCode (FK), Label, Unit (Pct/Count/GWh), ExtraDimCategory	metric_definitions	One row per KPI. Unit determines axis formatting in RDL. ExtraDimCategory = AgeBucket, ObligationType, EUC, etc.
LookupValue	LookupId, Category, Code, Label, SortOrder	lookup_values	Single multi-category lookup: ReasonCode (01–09), AgeBucket (a/b/c), ObligationType, YearBand, RejectionCode, MRECode. Replaces 6 tiny tables.

Entity / Table	Key Columns	Maps To	Why It Exists
MetricValue	Id, ShipperShortCode, ShipperId?, ReportCode, MetricCode, ReportingPeriod, ProductClassCode, Value, IngestionFileId	metric_values	Central fact table (~95% of all KPIs). ShipperId nullable = industry total row. Composite index on (ReportingPeriod, ShipperShortCode, MetricKey) powers all SP queries.
AqCorrectionByReason	Id, ShipperId, PeriodId, ReasonCodeLookupId, MprnCount	aq_corrections_by_reason	Dedicated fact for report 2A.8 (AQ Correction by Reason Code). 8 reason codes × 12 months × N shippers.
SupplyPointSnapshot	Id, GasDay, ShipperId, ClassId, EucCode, LdzCode, MprnCount, AqRoll	supply_point_snapshots	Finest-grain fact from 202604_SupplyPointCounts (Type-D file). Enables recalculation of industry aggregates and LDZ breakdowns.
IngestionFile	Id, FileName, SourceSystem, FileType, Status, ValidationStatus, BlobPath, FileHash, IngestionJobId	ingestion_files	Audit trail per processed file. FileHash deduplicates re-submissions. Links ValidationErrors and MetricValues back to the source file.
IngestionJob	Id, JobName, ReportingPeriod, Status, TriggeredBy, CorrelationId, ParentJobId	ingestion_jobs	One job per pipeline run (month). ParentJobId supports retry lineage. CorrelationId threads all logs and SignalR events.
ValidationError	Id, IngestionFileId, ErrorCode, ErrorMessage, ColumnName, OriginalValue, Severity	validation_errors	Per-file, per-rule failure details. Used to build the notification HTML table (first 10 rows per rule as required by US6).
Report	Id, ReportTypeId, ReportingPeriod, Status, FilePath_PDF/Excel/PPTX, ObservationsText, ObservationsBy	reports	Represents a published output report. Stores blob paths for PPTX/XLSX/PDF exports and analyst observations.
ReportType	Id, Code (SCH2A/SCH2B), ScheduleRef, Label, Audience	report_types	SCH2A = Anonymised (Industry), SCH2B = Non-Anonymised (PAC). Audience drives permission checks (reports.anonymised.view vs reports.nonanonymised.view).
FactReadPerformance	ReportMonth, ShipperCode, ProductClass, ReadPerfPct, EstimatedPct, CheckReadCount, TotalSites, IsCompliant	fact_read_performance (VIEW)	SQL View – no physical table. Aggregates MetricValue rows for 2A.1 and 2A.5 into a Power BI-friendly shape. No ETL writes here.
ShipperProductClass	ShipperId + ProductClassId + ReportingPeriod (composite PK), EstimatedPct,	shipper_product_classes	Denormalised snapshot per shipper × class × month. Computed by the ETL post-persist step for fast Power BI tabular queries without joining MetricValue.

Entity / Table	Key Columns	Maps To	Why It Exists
	ReadPerfPct, NoMeterCount		
PafaUser / Role / Permission	Standard RBAC triple: User → UserRole → Role → RolePermission → Permission	pafa_users/roles/permissions	Four roles: PafaUser (anon view), PafaAdmin (full), PacMember (non-anon view), Shipper (own anon). Permission codes gate API endpoints AND the AnonMode SP parameter.

2.2 EF Core Configuration Audit

Based on the current project files (IEntityTypeConfiguration classes inspected), the following table shows exactly what is already coded, what is missing, and what additions are needed on existing configs.

Orange background = MISSING configuration class	Green background = Add: (property/index to append)	White/grey = Already coded and correct
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Configuration Class	Critical Config Choices	Missing / To-Add vs Current Code
ShipperConfiguration	HasIndex(ShortCode).IsUnique(); Property(Name).HasMaxLength(150)	Add: Property(AnonymisedLabel).HasMaxLength(100); HasIndex(AnonymisedLabel).IsUnique() — needed for SP @AnonMode CASE switch and uniqueness guarantee
ShipperAliasConfiguration	HasIndex(ShipperId, IsActive); HasOne(Shipper).WithMany(ShipperAlias es)	Already coded. Verify ValidTo is nullable for active aliases. Add HasIndex(AliasCode, ValidFrom) for ETL resolution queries.
ReportTypeConfiguration	HasData(SCH2A, SCH2B); Audience enum → int	Already coded. Add: Property(IsAnonymised).HasDefaultValue(true) computed from Audience to simplify permission checks in API.
ProductClassConfiguration	ValueGeneratedNever(); HasData(PC1– PC4); AqThresholdLow → numeric(12,4)	Already coded. Confirm MinReadPercentage seed values: PC1=97.5, PC2=90.0, PC3=90.0, PC4-Monthly=90.0, PC4- Annual=20.0.
EucBandConfiguration (NEW)	ToTable("euc_bands"); PK = EucCode string; AqThresholdMinKWh + AqThresholdMaxKWh → bigint	MISSING — not yet in project. Needed for 2A.7, 2A.9, 2A.10, Rpt_1364 EUC dimension. Seed 9 rows: EUC01 (< 73,200 kWh) through EUC09 (≥ 58,600,000 kWh).
ReportDefinitionConfiguration (NEW)	ToTable("report_definitions"); PK = ReportCode string; RollingMonths int; DisplayOrder int	MISSING. 22 rows to seed (2A.1–2A.19 + 3 × 2B.x). Drives SP parameters and RDL DataSet queries. Without this, SP @RollingMonths must be hardcoded per report.
MetricDefinitionConfiguration (NEW)	ToTable("metric_definitions"); MetricCode string PK; Unit enum (Pct/Count/GWh); FK ReportCode	MISSING. ~40 MetricCode rows. Powers RDL axis formatting (% vs count vs GWh) and dataset column aliasing.
LookupValueConfiguration (NEW)	ToTable("lookup_values"); HasIndex(Category, Code).IsUnique(); SortOrder int	MISSING. 6 categories, 25 rows. Replaces ad-hoc string comparisons in ETL. ETL

Configuration Class	Critical Config Choices	Missing / To-Add vs Current Code
		resolves AgeBucket, ReasonCode, ObligationType to LookupId FK before insert.
MetricValueConfiguration	HasIndex(ReportingPeriod, ShipperShortCode, MetricKey); HasOne(Shipper) nullable; HasOne(IngestionFile)	Already coded. Add: Property(ReportCode).HasMaxLength(10); Property(EucCode).HasMaxLength(10); Property(LookupValueId) nullable — these extra columns allow direct SP join without sub-select.
AqCorrectionByReasonConfiguration (NEW)	ToTable("aq_corrections_by_reason"); HasOne(ReasonCode_LookupValue).WithMany()	MISSING. Dedicated fact for report 2A.8. Separating from MetricValue avoids a sparse pivot table for the 8 reason codes.
SupplyPointSnapshotConfiguration (NEW)	ToTable("supply_point_snapshots"); HasIndex(GasDay, ShipperId, ClassId, EucCode); AqRoll → bigint	MISSING. Stores the finest-grain Type-D file. GasDay = DateOnly. Essential for LDZ-level drill-down in Power BI.
ReportConfiguration	HasIndex(ReportingPeriod, ReportTypeId); FilePath_PDF/Excel/PPTX → varchar(1000)	Already coded. Add: Property(AnonymisedMode).HasDefaultValue(true) — tracks which version (2A vs 2B) the file represents. Add HasOne(IngestionJob) nullable FK.
ValidationErrorConfiguration	HasIndex(IngestionFileId); ErrorCode.HasMaxLength(50)	Already coded. Add: Property(LineNumber) int — needed to render the first 10 row examples in the notification email (US6 acceptance criteria).
IngestionJobConfiguration	HasIndex(ReportingPeriod); HasIndex(Status); HasOne(ParentJob).WithMany(RetryJobs)	Already coded. Add: Property(AnonMode) bit — records whether this job produced the anonymised or non-anonymised extract. Needed for Report FK linking.
FactReadPerformanceConfiguration	HasNoKey(); ToView("fact_read_performance")	Already coded as view. Ensure the SQL view definition is in a migration script. View should join MetricValue where MetricCode IN (EST_PCT, READ_PERF_PCT, CHECK_COUNT).
ShipperProductClassConfiguration	Composite PK (ShipperId, ProductClassId, ReportingPeriod)	Already coded. Add: Property(DataSource).HasMaxLength(20) — track whether row came from ETL or recalculated from SupplyPointSnapshot.

2.3 New Entity Classes to Create

These domain entity classes do not yet exist in the project and must be added to the PAFA.Domain layer before creating their EF configurations.

EucBand.cs

```
public class EucBand
{
    public string EucCode { get; set; } = null!; // "EUC01" .. "EUC09"
    public string Description { get; set; } = null!;
    public long AqThresholdMinKwh { get; set; }
    public long AqThresholdMaxKwh { get; set; }
}
```

ReportDefinition.cs

```

public class ReportDefinition
{
    public string ReportCode { get; set; } = null!; // "2A.1" .. "2A.19"
    public string Topic { get; set; } = null!;
    public string? SplitBy { get; set; }
    public string? ReportFormat { get; set; }
    public int RollingMonths { get; set; } = 12;
    public int DisplayOrder { get; set; }
    public ICollection<MetricDefinition> Metrics { get; set; } = [];
}

```

MetricDefinition.cs

```

public class MetricDefinition
{
    public string MetricCode { get; set; } = null!;
    public string ReportCode { get; set; } = null!; // FK
    public string Label { get; set; } = null!;
    public string Unit { get; set; } = null!; // "Pct" | "Count" | "GWh"
    public string? ExtraDimCategory { get; set; } // "AgeBucket" | "EUC" | "ObligationType"
    public ReportDefinition Report { get; set; } = null!;
}

```

LookupValue.cs

```

public class LookupValue
{
    public int LookupId { get; set; }
    public string Category { get; set; } = null!; // "ReasonCode" | "AgeBucket" | ...
    public string Code { get; set; } = null!;
    public string Label { get; set; } = null!;
    public int SortOrder { get; set; }
}

```

AqCorrectionByReason.cs

```

public class AqCorrectionByReason
{
    public Guid Id { get; set; }
    public Guid? ShipperId { get; set; }
    public int PeriodId { get; set; } // YYYYMM
    public int ReasonCodeLookupId { get; set; }
    public int MprnCount { get; set; }
    public Guid IngestionFileId { get; set; }
    public Shipper? Shipper { get; set; }
    public LookupValue ReasonCode { get; set; } = null!;
}

```

SupplyPointSnapshot.cs

```

public class SupplyPointSnapshot
{
    public Guid Id { get; set; }
    public DateOnly GasDay { get; set; }
    public Guid? ShipperId { get; set; }
    public int? ClassId { get; set; }
    public string? EucCode { get; set; }
    public string? LdzCode { get; set; }
    public int MprnCount { get; set; }
    public long AqRoll { get; set; }
    public Shipper? Shipper { get; set; }
}

```

2.4 ParrDbContext Additions

Add the following DbSet properties to the existing ParrDbContext / PafaDbContext:

```
// Add to existing DbContext:
public DbSet<EucBand> EucBands => Set<EucBand>();
public DbSet<ReportDefinition> ReportDefinitions => Set<ReportDefinition>();
public DbSet<MetricDefinition> MetricDefinitions => Set<MetricDefinition>();
public DbSet<LookupValue> LookupValues => Set<LookupValue>();
public DbSet<AqCorrectionByReason> AqCorrectionsByReason => Set<AqCorrectionByReason>();
public DbSet<SupplyPointSnapshot> SupplyPointSnapshots => Set<SupplyPointSnapshot>();
```

3. Stored Procedures for Power BI Report Builder

Power BI Report Builder connects to each stored procedure as a **DataSet**. The `@AnonMode bit` parameter is the single switch that drives both the anonymised (2A.x) and non-anonymised (2B.x) outputs from the same RDL file. The controller/user JWT claims map directly to this parameter.

Stored Procedure	Parameters	What Power BI Report Builder Uses It For
usp_GetKpiTimeSeries	@ReportCode nvarchar(10), @PeriodMonth int (YYYYMM), @RollingMonths int=12, @ClassId int?, @EucCode nvarchar(10)?, @MetricCode nvarchar(50)?, @AnonMode bit=1	Main line-chart/bar-chart dataset for ~18 of the 22 reports. Returns ShipperDisplay (anon or real), MonthLabel, MetricCode, Value ordered by Shipper × Period.
usp_GetIndustryKpi	@ReportCode, @PeriodMonth, @MetricCode?, @AnonMode bit	KPI card TextBoxes: returns CurrentValue, MonthlyChange, AnnualChange for industry total (ShipperId IS NULL). Used in every report header KPI box.
usp_GetTopShippers	@ReportCode, @PeriodMonth, @Top int=5, @Direction nvarchar(4)=ASC, @AnonMode bit	Highest/Poorest Shippers Tablix band at bottom of each slide. Returns ranked list with Value and delta.
usp_Get2A8_AqCorrection	@PeriodMonth, @RollingMonths=12, @AnonMode bit	Report 2A.8 only. Returns Fact_AqCorrectionByReason joined to LookupValue for ReasonCode label × 12 months × shipper. Used by Bar Chart grouped on ReasonCode.
usp_Get2A7_NoReads	@PeriodMonth, @ClassId int, @AnonMode bit	Report 2A.7: 4 Bar Charts (one per class). Returns Pct × EUC × YearBand from MetricValue where MetricCode IN (NO_READ_1YR..4YR). RDL groups series by YearBand.
usp_Get2A11_Class1Threshold	@PeriodMonth, @AnonMode bit	Report 2A.11a/b. Returns CLASS1_ABOVE_COUNT, CLASS1_ABOVE_AQ and RECLASSIFIED_COUNT × Shipper × Month. Two datasets on same page.
usp_Get2A12_AqReadPerf	@PeriodMonth, @ObligationType nvarchar(30), @AnonMode bit	Report 2A.12a/b/c. Filters MetricValue by MetricCode=AQ_READ_PERF_PCT and LookupValue.Code = @ObligationType. Returns bar chart data per shipper.
usp_GetSupplyPointSnapshot	@GasDay date, @ClassId int?, @EucCode nvarchar(10)?	Optional deep-drill: returns Fact_SupplyPointSnapshot aggregate for LDZ breakdown table. Not used in standard 22 reports but available for PAC ad-hoc queries.
usp_Get2A14_EnergyTheft	@PeriodMonth, @RollingMonths=12, @PeriodCode nvarchar(4)=P41, @AnonMode bit	Report 2A.14. Filters THEFT_* MetricCodes + LookupValue PeriodCode to distinguish P41 vs P106. Returns two paired series per month.
usp_GetAnonMapping	(no params)	Admin SP: returns full Shipper table with AnonymisedLabel ↔ ShortCode mapping.

Stored Procedure	Parameters	What Power BI Report Builder Uses It For
		Used by Power BI for Dim_Shipper dimension table dataset (disconnected slice).

3.1 Generic SP Template (usp_GetKpiTimeSeries)

This single SP handles ~80% of all report datasets. Copy and adapt for specialised cases.

```
CREATE OR ALTER PROCEDURE rpt.usp_GetKpiTimeSeries
    @ReportCode    nvarchar(10),
    @PeriodMonth   int,          -- YYYYMM of most recent month
    @RollingMonths int          = 12,
    @ClassId       int          = NULL,
    @EucCode       nvarchar(10) = NULL,
    @MetricCode    nvarchar(50) = NULL,
    @LookupCode    nvarchar(30) = NULL, -- e.g. ObligationType, AgeBucket
    @AnonMode      bit          = 1
AS
BEGIN
    SET NOCOUNT ON;

    -- Compute window start from Dim_Period (handles year-boundary correctly)
    DECLARE @MinPeriod int;
    SELECT @MinPeriod = MIN(p2.PeriodId)
    FROM (SELECT PeriodId FROM report_definitions rd
          JOIN (SELECT TOP (@RollingMonths) PeriodId FROM reporting_periods
                WHERE PeriodId <= @PeriodMonth ORDER BY PeriodId DESC) p
          ON 1=1) p2;

    SELECT
        CASE WHEN @AnonMode = 1
            THEN ISNULL(s.AnonymisedLabel, 'Industry Total')
            ELSE ISNULL(s.Name, 'Industry Total')
        END AS ShipperDisplay,
        s.ShortCode,
        p.PeriodId,
        p.MonthLabel,
        f.MetricKey      AS MetricCode,
        f.Value
    FROM metric_values f
    LEFT JOIN shippers s      ON s.Id      = f.ShipperId
    LEFT JOIN reporting_periods p ON p.PeriodId = f.ReportingPeriod
    LEFT JOIN lookup_values lv  ON lv.LookupId = f.LookupValueId
    WHERE f.ReportCode = @ReportCode
        AND f.ReportingPeriod BETWEEN @MinPeriod AND @PeriodMonth
        AND (@ClassId IS NULL OR f.ProductClassCode = CONCAT('PC', @ClassId))
        AND (@EucCode IS NULL OR f.EucCode = @EucCode)
        AND (@MetricCode IS NULL OR f.MetricKey = @MetricCode)
        AND (@LookupCode IS NULL OR lv.Code = @LookupCode)
    ORDER BY f.ShipperId, f.ReportingPeriod, f.MetricKey;
END
```

3.2 AnonMode Integration with JWT Claims

The API controller reads the user's role from the JWT and sets `@AnonMode` automatically. Shippers and PafaUsers always get anonymised mode; PacMembers and PafaAdmins get non-anonymised mode when they explicitly request 2B reports.

```
// In the Reports API controller:
private bool GetAnonMode(ClaimsPrincipal user)
{
    if (user.IsInRole("PafaAdmin") || user.IsInRole("PacMember"))
        return false; // AnonMode = 0 → real names
    return true;      // AnonMode = 1 → city names
}
```

```
}  
  
// Pass to SP call:  
cmd.Parameters.AddWithValue("@AnonMode", GetAnonMode(User) ? 1 : 0);
```

4. Power BI Report Builder — RDL Architecture

Each of the 22 reports is a standalone `.rdl` file embedded as a **Subreport** inside a single `Deck.rdl` master file. One `PageBreak` per sub-report produces **1 slide per report in PPTX export** and **1 tab per report in XLSX export**.

4.1 Report Page Layout Template

Page Size	13.333 in wide × 7.5 in tall — exactly 16:9. Set in Deck.rdl Report Properties. All margins = 0 in.
------------------	---

Each page is divided into 3 zones:

- Left zone (40% = 5.333 in): KPI TextBoxes — CurrentValue, MonthlyChange (M/M), AnnualChange (Y/Y). Color expressions: red for worsening, green for improvement.
- Top-right zone (60% wide, 55% tall): Primary chart — Line or Bar depending on report.
- Bottom-right zone (60% wide, 45% tall): Secondary chart or density (CDF) curve.
- Full-width bottom band (100% wide, 20% tall): Tablix "Highest/Poorest Shippers" + Observations TextBox.

4.2 Report Layout Reference per RDL

Code	Left Zone (40%) — KPI Cards	Top Right (60%×55%) — Main Chart	Bottom Right (60%×45%) — Secondary	Tablix Band (full width 20%)
2A.1	Industry EST_PCT ↑↓ + 2 deltas (M/M, Y/Y)	Line Chart: 12 months × all shippers, EST_PCT PC1	Line Chart: CHECK_COUNT PC2	Top 5 highest EST_PCT shippers
2A.2	Industry NO METER PCT ↑↓	Bar Chart grouped: 4 classes × 12 shippers	(none)	Top highest per class
2A.4	Industry TRANSFER_READ_PCT + delta	Line Chart: 12 months × shippers	Density CDF chart (XY Scatter, 12m distribution)	Poorest 5 shippers
2A.5	Industry avg READ_PERF_PCT per class	5 Bar Charts stacked: PC1/PC2/PC3/PC4M/PC4A per shipper	(none)	Poorest 5 per class
2A.7	Industry avg % no-read × YearBand	Bar Chart: EUC01–09 on X, 4 year bands as series (PC1)	Same chart for PC2/PC3/PC4 (pagination)	Top shippers with most no-reads
2A.8	Total corrections count + delta	Bar Chart grouped: 8 reason codes × 12 months	(none)	Shippers with highest corrections per reason
2A.9	STD_CF_COUNT total + delta	Line Chart: EUC01–09 × 12 months (EUC04 = secondary axis)	(none)	Top EUC04 shippers
2A.12a	Industry AQ_READ_PERF_PCT >=293k + delta	Bar Chart: shippers, ObligationType = MONTHLY_293K	Density CDF 12m	Poorest 5
2A.13	AQ_AT_RISK_PCT per class	Bar Chart grouped: 6 obligation categories on X	(none)	Top shippers by AQ at risk GWh

Code	Left Zone (40%) — KPI Cards	Top Right (60%×55%) — Main Chart	Bottom Right (60%×45%) — Secondary	Tablix Band (full width 20%)
2A.14	Theft claim count if any	TextBox "No data in this period" — conditional on =Count()=0	(none)	N/A
2A.16	Industry MIN_PCT vs 50% threshold	Line Chart: PC2 vs PC3 on same axes × 12 months	ReferenceLine at 50%	Non-compliant shippers
2A.18	COMR_COUNT + reject pct	Combo: Bar (Count) + Line (%) dual axis × 12 months	(none)	Top rejection shippers
2A.19	VACANT_COUNT trend	Bar Chart: monthly counts × 12 months	Proportion chart	Shippers with most vacant sites

4.3 AnonMode Parameter in RDL

In Deck.rdl and every sub-report, declare one report parameter:

```
<ReportParameters>
  <ReportParameter Name="ReportMonth">
    <DataType>DateTime</DataType>
    <DefaultValue><Values><Value>=Now()</Value></Values></DefaultValue>
  </ReportParameter>
  <ReportParameter Name="AnonMode">
    <DataType>Boolean</DataType>
    <DefaultValue><Values><Value>True</Value></Values></DefaultValue>
    <Hidden>True</Hidden> <!-- hide from end users; set programmatically -->
  </ReportParameter>
</ReportParameters>

<!-- In each DataSet, pass to the SP: -->
<QueryParameters>
  <QueryParameter Name="@AnonMode">
    <Value>=IIF(Parameters!AnonMode.Value, 1, 0)</Value>
  </QueryParameter>
</QueryParameters>
```

4.4 Chart Color Expression for ↑↓ Indicators

In TextBox expressions for MonthlyChange and AnnualChange, apply conditional colour. The direction depends on whether a higher value is "good" or "bad" for that KPI:

```
<!-- For KPIs where LOWER is better (e.g. EST_PCT, NO_METER_PCT): -->
=IIF(Fields!MonthlyChange.Value >= 0, "#C00000", "#375623")

<!-- For KPIs where HIGHER is better (e.g. READ_PERF_PCT, TRANSFER_READ_PCT): -->
=IIF(Fields!MonthlyChange.Value >= 0, "#375623", "#C00000")

<!-- Arrow symbol -->
=IIF(Fields!MonthlyChange.Value >= 0, Chr(8593) & " ", Chr(8595) & " ")
& Format(Abs(Fields!MonthlyChange.Value), "0.00%")
```

5. Role-Based Access & Anonymous Mode

The existing RBAC model (PafaUser, PafaAdmin, PacMember, Shipper) already seeds permissions correctly. The key addition is wiring `reports.nonanonymised.view` to the SP call's `@AnonMode` parameter, so the same Report Builder endpoint serves both versions without code duplication.

Permission / Role	PafaUser	PafaAdmin	PacMember	Shipper	AnonMode SP param
reports.anonymised.view	✓	✓	✓	✓	1 (anon)
reports.nonanonymised.view	X	✓	✓	X	0 (real names)
reports.anonymised.edit	X	✓	X	X	1
reports.nonanonymised.edit	X	✓	X	X	0
reports.download	✓	✓	✓	✓	both
users.create	X	✓	X	X	n/a
users.delete	X	✓	X	X	n/a

Important: Never add a second set of reports, views, or RDL files for the non-anonymised version. The single `Deck.rdl` handles both outputs via `@AnonMode`. The Shipper's own data (User Story 4 — "own data access") is further filtered by adding `AND s.ShortCode = @ShipperShortCode` in the SP when the caller's role is Shipper.

6. Export Pipeline (PPTX · XLSX · PDF)

6.1 Export Checklist

1. Confirm `metric_values` contains the target month (`PeriodId = YYYYMM`).
2. Run the ETL pipeline (`POST /api/pipeline/run`) and verify all 3 steps = Success.
3. Open `Deck.rdl` in Power BI Report Builder.
4. Set `@ReportMonth = target month`, `@AnonMode = True`.
5. Preview → check 5 spot slides (2A.1, 2A.5, 2A.8, 2A.13, 2A.19) vs PDF mock-up.
6. File → Export → PowerPoint → verify 22 slides, named by `PageName`.
7. File → Export → Excel → verify 22 tabs, Tablix data in native cells.
8. Repeat steps 4–7 with `@AnonMode = False` for the PAC (2B) version.
9. Compare industry totals between both exports — must be identical.

6.2 Automated Export via .NET HttpClient

```
public async Task<byte[]> ExportAsync(string format, bool anonMode, int year, int month)
{
    // format = "PPTX" | "EXCELOPENXML" | "PDF"
    var url = $"{_ssrsBase}/Deck.rdl?rs:Format={format}"
        + $"&ReportMonth={year}-{month:D2}-01"
        + $"&AnonMode={anonMode.ToString().ToLower()}";

    using var client = _httpClientFactory.CreateClient("SSRS");
    var bytes = await client.GetByteArrayAsync(url);

    // Store in blob and update Report.FilePath_* column
    var path = $"reports/{year}/{month:D2}/Deck_{(anonMode?"2A":"2B")}.{format.ToLower()}";
    await _blobStorage.UploadAsync(path, bytes);
    return bytes;
}
```

6.3 Axis-Rendering Warnings for PPTX

- Reports 2A.9 and 2A.10 use a secondary Y-axis (EUC04 on right). Test these first in Report Builder — the PPTX renderer may clip the secondary axis label. Workaround: set `SecondaryAxis.Maximum` to a fixed value if auto-scale fails.
- Reports 2A.4 and 2A.12a/b/c use a density/CDF XY Scatter chart. Well-supported in PPTX export but verify the data sort order (ascending Value) in the SP.
- Report 2A.18 is a Combo chart (Bar + Line). The Line series must be on `ChartType=Line` with `UseSecondaryAxis=True`. Validate the dual-axis alignment.
- Avoid PNG/EMF background images — they inflate the PPTX file size. Use solid fill colours instead.

7. User Story Coverage Map

The table below maps each acceptance criterion from the 6 user stories to the implementation component that satisfies it.

US	Acceptance Criteria	Satisfied By	Implementation Detail
US 1	Cron job checks files every hour in the 16th–20th window	PipelineBackgroundService + Scheduled Trigger	Add a Quartz.NET or IHostedService CronJob that calls IPipelineBackgroundService.EnqueueAsync on schedule. The current manual POST /api/pipeline/run becomes the fallback.
US 1	Validation block — file not processed until all checks pass	ParseAndValidateFilesCommand (Step 2)	Step 2 returns result.Success = false if any rule fails; SkipRemainingAsync prevents Step 3 (persist) from running.
US 1	Versioned filenames (filename_v1, filename_v2)	IngestionFile.FileHash + SharePointFileSource	FileHash (SHA-256) detects re-submissions. Versioned filename stored in IngestionFile.FileName. Previous versions remain with Status = ValidationFailed.
US 2	Manual trigger outside the schedule window	POST /api/pipeline/run	Already implemented. Accessible to PafaAdmin role. Returns 202 + jobId for SignalR tracking.
US 3	File naming convention: MOD520A__PAF_Reports_MMM26_*	SharePointFileSource validation	Add a regex check on FileName in Step 1 (ImportFilesCommand): pattern = ^MOD520A__PAF_Reports_[A-Za-z]{3}d{2}_ (Anonymised Non.Anonymised)\.xlsx\$. Log ValidationError if mismatch.
US 3	Folder structure: PAFA/YYYY/MM/	SharePointSettings.BasePath	Already in SharePointSettings. ETL builds path as \$"{BasePath}/{year}/{month:D2}/". Files outside this path are flagged with ErrorCode = FOLDER_STRUCTURE_INVALID.
US 4	Retry with exponential backoff on SharePoint failure	SharePointFileSource + IRemoteFileSource	Implement Polly retry policy: 3 retries at 5 min / 15 min / 30 min. After all retries fail, trigger ServiceBusNotificationService with failure type SHAREPOINT_UNAVAILABLE.
US 5	Quarantine invalid files; rest of pipeline continues	ParseAndValidateFilesCommand	Files with ValidationStatus = Failed are excluded from the importedFiles list passed to Step 3. The blob path stores them under a /quarantine/ prefix. The pipeline continues for all other files.
US 5	Correlation ID in all log entries	PipelineExecutionState.CorrelationId	Already generated in ProcessJobAsync as Guid.NewGuid(). Passed to all three MediatR commands. All log lines include [CorrelationId={correlationId}].
US 6	Consolidated failure notification after all files processed	ServiceBusNotificationService	Notification sent once at end of Step 2, not per file. ServiceBusNotificationService.SendAsync groups all failed IngestionFiles and their ValidationErrors.
US 6	First 10 row examples per failed rule in HTML table	ValidationError.RowNumber (to add)	ETL parsers must store the source row number in ValidationError.RowNumber. The notification service orders by RowNumber

US	Acceptance Criteria	Satisfied By	Implementation Detail
			and takes first 10 per ErrorCode, renders as HTML table.
US 6	Notification includes file name, month, source, failure type	ValidationNotification entity	Already mapped: FileName, ReportingPeriod, SourceSystem, Severity, ErrorMessage. Wire these fields into the HTML email template.

8. Implementation Roadmap

15-step sequence from EF configuration gaps through to dual-mode PPTX/XLSX export. Total estimated effort: 22–26 working days.

Step	Deliverable	Key Actions	Files / Classes Involved	Est. Days
1	Missing EF Configurations	Create EucBandConfiguration, ReportDefinitionConfiguration, MetricDefinitionConfiguration, LookupValueConfiguration; seed all static data	New: EucBandConfiguration.cs, ReportDefinitionConfiguration.cs, MetricDefinitionConfiguration.cs, LookupValueConfiguration.cs	1
2	Update existing configs	Add AnonymisedLabel to ShipperConfiguration; add EucCode + ReportCode + LookupValueId to MetricValueConfiguration; add RowNumber to ValidationErrorConfiguration	ShipperConfiguration.cs, MetricValueConfiguration.cs, ValidationErrorConfiguration.cs	0.5
3	New fact configurations	Create AqCorrectionByReasonConfiguration and SupplyPointSnapshotConfiguration; add entity classes to domain project	New: AqCorrectionByReasonConfiguration.cs, SupplyPointSnapshotConfiguration.cs	0.5
4	EF Core migration	dotnet ef migrations add AddParrReportingLayer; apply to DB; verify all indexes created	ParrDbContext.cs, Data/Migrations/	0.5
5	ETL parsers — Flat files	Parse: Read_Performance_by_Shipper_5, Shipper_Transfer, Class_3_conversion, Supply_Points files; direct long load into MetricValue	New: FlatFileParser.cs in PAFA.Extraction	2
6	ETL parsers — Matrix Wide files	Parse: MOD520A all sheets, Report_1A/1B/2/3, 2B_21, Confirmed_Energy_Theft; skip-row logic + unpivot; anonymous label → ShortCode mapping	New: MatrixFileParser.cs, UnpivotHelper.cs	3
7	ETL parsers — Multi-axis files	Parse: AQ_at_Risk (class unpivot), Rpt_1364 (EUC × shipper × 8 sheets), EUC09_Reporting (multi-header), 202604_SupplyPointCounts	New: MultiAxisParser.cs; extend SupplyPointSnapshotRepository	2
8	Stored Procedures (SQL)	Create all 10 SPs in migration script; parameterise @AnonMode; test with EXEC usp_GetKpiTimeSeries N'2A.1', 202604, 12, NULL, NULL, NULL, 1	New SQL migration: AddReportingStoredProcedures.sql	2
9	SQL View: fact_read_performance	Write CREATE OR REPLACE VIEW joining metric_values for EST_PCT + READ_PERF_PCT; verify FactReadPerformanceConfiguration maps correctly	New SQL migration or Fluent API HasAnnotation	0.5
10	Power BI Report Builder — Datasets	Create shared DataSets (one per SP) in Report Builder; test	Report Builder project (.rptproj)	1

Step	Deliverable	Key Actions	Files / Classes Involved	Est. Days
		connection string; verify @AnonMode parameter toggle		
11	RDL unitaires (22 reports)	Build 22 .rdl files: charts, TextBox KPI, Tablix observations; use PageSize 13.333×7.5in; parameterise @AnonMode per subreport	Reports/Report_2A1_*.rdl through Report_2A19_*.rdl	7
12	Deck.rdl (master)	Assemble 22 sub-reports into single Deck.rdl; PageBreak between each; verify PageName = report code	Deck.rdl	1
13	PPTX / XLSX export test	Export Deck.rdl → PPTX (22 slides); XLSX (22 tabs); compare slide-by-slide to PDF mock-up; validate chart rendering on axes 2A.9/2A.10/2A.18	Manual testing in Report Builder	1
14	Dual-mode validation	@AnonMode=1 → city names; @AnonMode=0 → real names; industry totals must be identical in both outputs	usp_GetIndustryKpi + cross-check script	0.5
15	Notification email (US6)	Complete ValidationError.RowNumber population; wire ServiceBusNotificationService to render HTML table of first 10 failed rows per rule	ServiceBusNotificationService.cs, ValidationError entity	1

8.1 Priority Order

Week 1 (Days 1–5): Steps 1–4 (new EF configs + migration) + Step 5 (flat-file parsers). Goal: metric_values contains real data for at least 3 files.

Week 2 (Days 6–10): Steps 6–8 (all parsers + stored procedures). Goal: all 24 files parsed and all 10 SPs executable in SQL Server Management Studio.

Week 3 (Days 11–17): Steps 9–12 (SQL view + Report Builder RDL files + Deck.rdl). Goal: Deck.rdl renders all 22 reports in Preview.

Week 4 (Days 18–22): Steps 13–15 (export validation + dual-mode + US6 notification). Goal: both PPTX exports (anon + non-anon) delivered and validated against the PDF mock-up.